PARIMAL GAVALI

MSc student | Applied Machine Learning and Software Engineering

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ABOUT ME

Master's student in Informatics for Natural Sciences with nearly three years of backend development experience at Accenture for Commerzbank. Project experience in agricultural remote sensing, crop monitoring and time-series machine learning.

WORK EXPERIENCE

06/2026 - present

Student Assistant - Web Development

Bielefeld University | Bielefeld, Germany | Part-time

- Develop and maintain BIIGLE, a web-based scientific application, using Vue.js and Laravel.
- Improve user-interface components and backend APIs; investigate defects, implement fixes and participate in Git-based code reviews.

03/2023 - 09/2024

Application Development Analyst

Accenture Solutions | Pune, India | Client: Commerzbank

- Developed and maintained Java and Spring Boot microservices for banking workflows and internal applications.
- Translated business requirements into backend solutions with business analysts, product owners and developers.
- Implemented unit and integration tests using JUnit and Mockito, maintaining test coverage above 90%.
- Investigated production incidents and implemented fixes; reviewed code and resolved SonarQube findings.

12/2021 - 02/2023

Application Development Associate

Accenture Solutions | Pune, India | Client: Commerzbank

- Developed Java and Spring Boot backend components and REST services for enterprise banking applications.
- Wrote automated tests, investigated application logs and resolved software defects.

EDUCATION AND TRAINING

10/2024 - present

M.Sc. Informatics for Natural Sciences

Bielefeld University | Bielefeld, Germany

- Expected graduation: May-June 2027.
- Relevant coursework: Machine Learning, Deep Learning, Data Mining, Big Data Analytics and Statistical Natural Language Processing.

08/2025 - 01/2026

Erasmus+ Exchange Semester - Computer Science

University of Tartu | Tartu, Estonia

- Relevant coursework: Statistical Machine Learning, Applied Cryptography and Databases.

08/2017 - 08/2021

Bachelor of Engineering - Electronics and Telecommunications

Savitribai Phule Pune University | Pune, India

PROJECTS

10/2026 - 11/2026

Crop-Stress Early-Warning System

GitHub

Personal project | Python, GeoPandas, LightGBM, FastAPI, PostGIS

- Built a geospatial pipeline combining Sentinel-1/2 satellite observations and DWD weather for winter-wheat monitoring in North Rhine-Westphalia.
- Modelled 7-, 14 and 21-day risks of peer-relative canopy anomalies, preserving observation masks and time since each measurement.
- Compared logistic regression, LightGBM and GRU-D/LSTM models using season-based splits; the calibrated logistic model reached 0.80 AUROC at 14 days on held-out 2024 data.
- Implemented a FastAPI/PostGIS service and React/MapLibre dashboard with interpretable risk explanations, season replay and English/German read-aloud.

04/2026 - 08/2026

Early Sepsis Prediction from ICU Time-Series Data

GitHub

Bielefeld University | Python, PyTorch, scikit-learn, MIMIC-IV

- Developed a machine-learning pipeline for early sepsis prediction using longitudinal intensive care unit (ICU) data.
- Trained and compared Random Forest, gated recurrent unit (GRU) and long short-term memory (LSTM) models across prediction horizons up to 25 hours before sepsis onset.
- Engineered temporal features using fixed and expanding observation windows.
- Evaluated class-imbalanced predictions using area under the ROC and precision-recall curves (AUROC/AUPRC), sensitivity and specificity.

DIGITAL SKILLS

Programming	Python, Java, SQL
Backend and web	Spring Boot, FastAPI, Laravel, Vue.js, React, REST APIs, microservices
ML and data	PyTorch, scikit-learn, LightGBM, TensorFlow, pandas, NumPy, time-series analysis
Remote sensing / GIS	Sentinel-1/2, GeoPandas, Rasterio, MapLibre
Applied AI	OpenAI API, LangChain, retrieval-augmented generation (RAG), vector databases
Testing and quality	JUnit, Mockito, SonarQube, Postman
Development tools	Docker, Git, GitHub, Bitbucket Pipelines
Databases	PostgreSQL, PostGIS, Oracle

LANGUAGE SKILLS

English: C1 (advanced) | German: A2.1 (elementary)

HACKATHONS

16/10/2026 - 18/10/2026

Nebius x NVIDIA Hackathon

GitHub

Siemens Supplier Risk Intelligence

- Contributed to a team prototype for supplier-risk research, screening and comparison.

CERTIFICATIONS

Nebius Agentic AI Builder Certification
Nebius Academy
View credential